



Please find data sources for this research question:

Which audio and metadata features best predict a song's commercial success on Spotify?

Based on my research, I now have comprehensive information about data sources for the research question on Spotify commercial success prediction. Let me compile this into a detailed report.

Data Sources for Predicting Song Commercial Success on Spotify

Primary Research Papers and Studies

Recent Academic Research:

The most directly relevant recent work is "Prediction of Spotify Chart Success Using Audio and Streaming Features" by Cabansag and Ntegeka (May 2025). This study analyzed 14,639 unique songs from the 2024 U.S. Top 200 Spotify Daily Charts using the Spotify Web API, implementing a classification pipeline with Random Forest and XGBoost models achieving 97% accuracy and macro F1-scores near 0.95. The research found that even audio-only models without stream counts retained above 89% predictive accuracy.^[1]

Another comprehensive study, "A Study For ML-Based Song Hit Prediction Using Spotify" by Dimolitsas et al. (2023), gathered nearly 18,500 hit and non-hit songs and achieved approximately 86% accuracy predicting Billboard success using Random Forest and Support Vector Machine algorithms.^[2]

"Decoding Spotify Hits: Statistical and Predictive Analysis of Track Features Driving Song Popularity" (2025) employed regression and classification machine learning algorithms on approximately 27,000 songs from Billboard's Hot 100 charts, identifying loudness, instrumentality, acousticness, duration, and speechiness as top feature predictors.^[3]

Saragih's 2023 study on "Predicting song popularity based on Spotify's audio features" analyzed 92,755 songs and identified the most important features as release date, loudness, instrumentality, acousticness, duration, speechiness, energy, valence, danceability, and liveness.^[4]

Key Feature Findings:

Research consistently identifies specific audio features as significant predictors. A study analyzing top tracks found that danceability, energy, and valence were moderately correlated and strongly associated with streaming success. The research on Billboard Hot 100 prediction identified that alongside streaming popularity, instrumentality, valence, duration, and

speechiness provided the strongest predictive influence with 90%+ accuracy across multiple algorithms.^[5] ^[6]

Available Datasets

Large-Scale Public Datasets:

HuggingFace Spotify Tracks Dataset (114,000 tracks across 125 genres) - This is the most accessible starting point for researchers, offering 125 different music genres with associated audio features in CSV format. The dataset has been downloaded over 2,000 times and includes comprehensive audio features.^[7]

Kaggle Spotify 1 Million Tracks Dataset - Consists of approximately 1 million tracks with 19 features extracted between 2000-2023, including 61,445 unique artists and 82 genres.^[8]

Spotify Million Playlist Dataset (MPD) - Available through Zenodo and Alcrowd, this dataset contains 1 million user-created playlists with 3 million unique tracks, 3 million albums, and 1.3 million artists from 2010-2017. The dataset includes playlist metadata and track-artist-playlist associations suitable for collaborative filtering research.^[9]

Kaggle Spotify Tracks Attributes and Popularity Dataset - Features detailed metadata (artist names, album names, track names) alongside 16+ audio features including popularity scores, ideal for exploratory analysis and feature engineering.^[10]

Music Streaming Sessions Dataset (MSSD) - Spotify Research released 150 million listening sessions with approximately 3.7 million unique tracks' audio features and metadata, representing the largest collection of such track metadata publicly available.^[11]

Audio Feature Specifications

The Spotify Web API provides standardized audio features including:^[12]

- **Danceability** (0.0-1.0): Suitability for dancing based on tempo, rhythm, and beat strength
- **Energy** (0.0-1.0): Intensity and activity perception
- **Acousticness** (0.0-1.0): Confidence measure of acoustic nature
- **Valence** (0.0-1.0): Musical positivity (sad to happy)
- **Instrumentalness** (0.0-1.0): Likelihood of no vocal content
- **Speechiness** (0.0-1.0): Presence of spoken words
- **Loudness** (dB): Overall track loudness
- **Tempo** (BPM): Estimated beats per minute
- **Key**: Musical key (0-11)
- **Mode**: Major (1) or minor (0)
- **Duration** (milliseconds): Track length
- **Explicit**: Binary indicator of explicit content
- **Liveness** (0.0-1.0): Audience presence in recording

- **Time Signature:** Time signature (e.g., 4/4)

API and Data Access Methods

Spotify Web API - The primary official source for audio features and metadata, requiring OAuth authentication. Developers can request individual track audio features through the endpoint documentation.^[12]

Spotipy Python Library - A Python wrapper for the Spotify Web API that simplifies data collection for research purposes.^[13]

Academic Datasets with Extended Context

Free Music Archive (FMA) Dataset - Research comparing FMA training data with Spotify test data identified speechiness, valence, instrumentality, and acousticness as most effective features for predicting popularity across genres.^[14]

Million Song Dataset - A freely-available collection of audio features and metadata for one million contemporary popular music tracks provided by The Echo Nest, now integrated with complementary datasets including Last.fm tags, musiXmatch lyrics, and cover song information.^[15]

Research-Specific Data Collections

Top Artists Dataset (2023) - Recent research leveraged Convolutional Neural Networks on Spotify's top 100 artists data from February 2023, achieving 97% F1 scores in popularity prediction.^[16]

Last.fm **Dataset** - For social network analysis in hit prediction: 2.7 million users, 10 million songs, 300 million plays demonstrating that social influence improved hit song prediction accuracy from 14% to 21%.^[17]

Ethical and Access Considerations

Researchers should note that while Spotify audio features are freely available through the API, republishing extensive Spotify audio feature datasets as standalone resources may violate Spotify's Developer Terms of Service. However, derivatives used for academic research with proper attribution are generally permitted in peer-reviewed publications.^[18]

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Song Popularity Prediction Using FMA Dataset, UC Berkeley^[14]

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Spotify Community Discussion on Publishing Audio Features Datasets^[18]

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